Arjun Sethi-Olowin

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Education

University of Washington September 2023–Present
Ph.D. candidate in Applied Mathematics
GPA: 3.7
Advanced to candidacy: June 2026

Rice University August 2020–May 2023
B.A. in Applied Mathematics and Mathematics
GPA: 3.89

University of Illinois at Urbana-Champaign August 2019–May 2020

Research Experience

Graduate Research Intern at Sandia National Labs June 2026–September 2026
Continued collaboration with Bobby Baraldi at Sandia National Labs' Rapid Optimization Library team, developing proximal operator approximations for non-smooth trust-region optimization.
Results to appear in the 2026 CSRI Summer Proceedings.

Graduate Research Intern at Sandia National Labs June 2025–September 2025
Collaborated with Bobby Baraldi of Sandia National Labs' Rapid Optimization Library team, implementing and theoretically guaranteeing mixed-precision gradient methods for trust-region algorithms in PDE-constrained optimization.
Results published in the 2025 CSRI Summer Proceedings.

Graduate Research on Structured Matrices June 2024–Present
Conducting research under the supervision of Dr. Heather Wilber, focusing on the development and analysis of efficient algorithms for solving linear systems involving highly structured matrices.
Developing publicly accessible MATLAB classes designed to enhance numerical efficiency and robustness in computations with such structured matrix classes.

Graduate Research on Kalman Smoothing and Filtering April 2024–Present
Conducting research under the supervision of Dr. Aleksandr Aravkin on the development and theoretical analysis of a hybrid Kalman filtering/smoothing method for systems subject to non-Gaussian error distributions.

Prosthetic Hand Senior Design Research Project August 2022–May 2023
Built a prosthetic hand using 3D printing and used machine learning and computer vision to control the hand based on physical touch sensors and a mounted camera.
Presented my work to the Rice Computation and Applied Mathematics department upon completion.

Parameter Estimation Research Project August 2022–May 2023
Worked with Professor Matthias Heinkenschloss and his research group.
Implemented the 2 compartment Pinsky-Rinzel neuron model for parameter estimation using Pyomo in Python.

Emory REU - Models Meet Data Summer 2022
Undertook a research project on optimal control theory and solved HJB equations with neural networks under the supervision of Lars Ruthotto.
Presented my work to the Emory Mathematics Department.

PDE and Inverse Problems Directed Reading January 2022–May 2022
Studied finite difference method for partial differential equations with varying time-stepping methods on inverse problems under Dr. Mario Bencomo and Dr. Jesse Chan.

Work Experience

Teaching Assistant: Optimization Methods in Finance September 2026–December 2026
Host office hours, proctor exams and grade homework and exams for CFRM 507, a course on optimization methods for financial modeling.

Teaching Assistant: Numerical Linear Algebra September 2025–December 2025
Host office hours, proctor exams and grade homework and exams for AMATH 584, an introductory graduate level course on numerical linear algebra.

Teaching Assistant: Machine Learning for Finance March 2025–June 2025
Host office hours, proctor exams and grade homework and exams for CFRM 521/421, a course on machine learning with application to finance.

Teaching Assistant: Quantitative Risk Management January 2025–March 2025
Host office hours, proctor exams and grade homework and exams for CFRM 541, a Masters level computational finance course on quantitative risk management.

Teaching Assistant: Credit Risk Management September 2024–December 2024
Hosted office hours, proctored exams and graded homework and exams for CFRM 542, a Masters level computational finance course on credit risk management.

Teaching Assistant: Calculus September 2023–June 2024
Prepared and led four weekly lecture sections for undergraduate Calculus I.
Hosted office hours, proctored exams and graded homework and exams.

Rice Learning Assistant January 2021–May 2023
Teaching assistant for the introductory applied mathematics course at Rice University.
Prepared and led weekly lectures, hosted office hours and graded homework and exams.

GIS Student Assistant January 2021–May 2023
Provided GIS support for ImagineRio, a searchable digital atlas that illustrates the social and urban evolution of Rio de Janeiro.

Developed and taught courses on GIS tools at Fondren Library to Rice students and the public.

Orientation Week Advisor August 2021–May 2022, August 2022–May 2023
Acted as a student resource to a group of incoming freshmen for orientation week as well as over the course of the entire school year including academic and well-being support.
Engaged in trainings on examples and responses to mental health struggles common in young adults entering college.

Publications

- Sethi-Olowin, A & Baraldi, B. Viscosity-based Proximal Operators in Nonsmooth, Regularized Optimization. Computer Science Research Institute Summer Proceedings 2026, Sandia National Laboratories, Albuquerque, NM & Livermore, CA (2026, forthcoming)
- Sethi-Olowin, A & Baraldi, B. Mixed Precision Linear Algebra in Trust-Region Methods for PDE Constrained Optimization (SAND Report No. SAND2025- 14267O). Computer Science Research Institute Summer Proceedings 2025, Sandia National Laboratories, Albuquerque, NM & Livermore, CA (2025)

Conferences

- "Efficient Solvers for Linear Optimization via the Hierarchically Semi-Separable Matrix Factorization." Invited minisymposium talk, SIAM Conference on Mathematics of Data Science, Salt Lake City, UT (November 2026, scheduled)
- "Efficient Solvers for Linear Optimization via the Hierarchically Semi-Separable Matrix Factorization." Invited minisymposium talk, Joint Mathematics Meetings, Chicago, IL (January 2027, scheduled)

Awards and Honors

- Graduated Cum Laude from Rice University May 2023
- Emory REU First Prize Poster July 2022
- UIUC Deans List Fall 2019

Skills

- Spoken languages: English, Hindi, Mandarin

- Programming languages: Python, Mathematica, Matlab, LaTeX
- Frameworks and technical skills: PyTorch, TensorFlow, High Performance Computing, ArcGIS

Extracurricular

- Memberships: UW SIAM Student Chapter, American Mathematical Society
- Service: UW Applied Mathematics Department Community Committee